Lesson 12 Combinatorial Proofs

In Lesson 11, we introduced basic concepts in counting such as the multiplication principle, permutation, and combination. We also did a combinatorial argument connecting permutation and combination to derive a formula for combination. In this lesson, we continue on **combinatorial arguments and proofs**.

Proposition 1 Let A be a finite set with $|A| = n$, then $|\mathcal{P}(A)| = 2^n$.

Proof The power set $\mathcal{P}(A)$ of A, is the set of all subsets of $A = \{a_1, \dots, a_n\}$. To count the subsets, we view a subset as a series of decisions whether to include a_1, a_2, etc., and use a n-element list $(b_1, b_2, \dots, b_n)$ to record: $b_k = 1$ if the subset includes a_k, and $b_k = 0$ otherwise. Each subset corresponds to a k-element *binary* list, where each element is 0 or 1, For example, $\{a_1, a_3, a_4\}$ corresponds to $(1, 0, 1, 1, 0, \dots, 0)$. The multiplication principle says that the number of such lists is $2 \times 2 \times \cdots \times 2 = 2^n$.

A combinatorial or counting proof is often an effective way to prove an identity. The above proof illustrates a general strategy of combinatorial proofs:

- Map a counting problem to one that we can do.

In this example, we connect subsets to binary lists, which we can count.

Proposition 2 Let n be a nonnegative integer, then we have the identity,

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Proof Let A be a set with n elements. As we have seen in the earlier proposition, the RHS (*right hand side*) is the number of subsets of A. We will show that the LHS (*left hand side*) also gives the number of subsets of A. By the definition of the combination $\binom{n}{k}$, A has $\binom{n}{0}$ 0-element subsets, $\binom{n}{1}$ 1-element subsets, $\binom{n}{2}$ 2-element subsets, etc. So, the number of all subsets of A is the sum:

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = \sum_{k=0}^{n} \binom{n}{k}$$

This example illustrates another general strategy of combinatorial proofs:

- Show that the LHS and the RHS are two ways of counting the same thing.

Binomial Theorem Let n be a non-negative integer. Then

$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$$

Note: Because of its appearance in the Binomial Theorem, the number $\binom{n}{k}$ is called a **binomial coefficient**.

What is the coefficient of x^3 in $(1-2x)^5$? By the Binomial Theorem,

$$(1-2x)^5 = \big(1+(-2x)\big)^5 = \sum_{k=0}^{5} \binom{5}{k} 1^{5-k}(-2x)^k = \sum_{k=0}^{5} \binom{5}{k} (-2x)^k$$

We have $n=5$, and x^3 corresponds to $k=3$, which is the following term,

$$\binom{5}{3}(-2x)^3 = \frac{5\times 4}{2\times 1}(-2)^3x^3 = -80x^3$$

The coefficient of x^3 is -80.

Before proving the Binomial Theorem, let's look at the binomial expansion for the first few values of n:

$$\begin{gathered}(x+y)^0 = 1\\ (x+y)^1 = x+y\\ (x+y)^2 = x^2+2xy+y^2\\ (x+y)^3 = x^3+3x^2y+3xy^2+y^3\\ (x+y)^4 = x^4+4x^3y+6x^2y^2+4xy^3+y^4\end{gathered}$$

The coefficients are $\binom{n}{k}$, which are numbers in the famous **Pascal's triangle**:

$$\begin{array}{ccccccccc} & & & & 1 & & & & \\ & & & 1 & & 1 & & & \\ & & 1 & & 2 & & 1 & & \\ & 1 & & 3 & & 3 & & 1 & \\ 1 & & 4 & & 6 & & 4 & & 1 \end{array} \qquad \begin{array}{ccccccccc} & & & & \binom{0}{0} & & & & \\ & & & \binom{1}{0} & & \binom{1}{1} & & & \\ & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & \\ & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & \\ \binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} \end{array}$$

We will give a combinatorial proof of the Binomial Theorem.

Proof The product $(x+y)^n = (x+y)(x+y)\cdots(x+y)$, when expanded, is the sum of all possible terms x^iy^j with x or y taken from each factor. Since each of the n factors contributes exactly one x or y in each x^iy^j term, the sum of the indices of each x^iy^j term is $i+j=n$: if there are k y's, then there are $(n-k)$ x's, resulting in $x^{n-k}y^k$. How many $x^{n-k}y^k$ terms are there? The answer is the number of ways you can choose k of the n terms to contribute a "y", and that is $\binom{n}{k}$. Finally, we sum from $k=0$ to n because there can be 0, 1, 2, …, or n y's in each term.